

Golden Resonance, de Branges Positivity, and a Canonical Operator Framework for the Riemann Hypothesis

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Abstract

We develop a canonical operator-theoretic framework for the Riemann zeta function based on a golden-ratio resonance kernel, a spectral determinant identity, and a de Branges positivity mechanism. The construction proceeds by a bounded, self-adjoint perturbation of the shift generator on $L^2(\mathbb{R})$, *introduced through a completely monotone, even kernel* $\kappa_{\varphi_g}(t) = \frac{1}{2\varphi_g} e^{-|t|/\varphi_g}$ *with* $\varphi_g = \frac{1+\sqrt{5}}{2}$. *Using a norm-resolvent limiting scheme and Birman–Kreĭn theory, we obtain :* (i) *a trace identity matching*

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1 Introduction

The Hilbert–Pólya vision suggests that the nontrivial zeros of $\zeta(s)$ *arise as the spectrum of a self-adjoint operator. Realizing this canonically — without tuning and with transparent arithmetic fingerprint — has remained elusive. We present a framework that aligns these demands by welding three pillars that mutually*

Golden resonance kernel. An even, completely monotone kernel fixed by geometric and analytic symmetry,

$$\kappa_{\varphi_g}(t) = \frac{1}{2\varphi_g} e^{-|t|/\varphi_g}, \quad \widehat{\kappa_{\varphi_g}}(\omega) = \frac{1}{1+\varphi_g^2\omega^2},$$

acting as a damping/modulation profile with rational spectral decay.

Spectral determinant identity. A zeta-regularized/Fredholm-type determinant $\Delta(z)$ *for a bounded self-adjoint perturbation of the shift generator encodes an arithmetic content through a Hadamard factorization locking* $\xi(\frac{1}{2}+iz)$ *upto an elementary factor.* **de Branges positivity.** *A Schur–class transfer function and its Herglotz–Biehler structure that imposes positivity identities forcing zeros onto the critical line.*

Guide to the paper. Section 2 fixes functional-analytic background. Section 3 builds the operator A via a golden-kernel perturbation and norm-resolvent limits. Section 4 proves the trace identity matching the explicit formula. Section 5 establishes the determinant identity and spectral bijection via Hadamard theory. Section 6 constructs the de Branges space and the positivity forcing critical-line localization. Appendices collect technical estimates, Birman–Kreĭn, and de Branges basics.

2 Preliminaries and notation

Notation 2.1. We work on $L^2(\mathbb{R})$ with the unitary Fourier transform $(\widehat{f})(\omega) = \int_{\mathbb{R}} f(t) e^{-i\omega t} dt$. Let $A_0 := -i \frac{d}{dt}$ with domain $H^1(\mathbb{R})$; on the Fourier side, A_0 is multiplication by ω . The golden kernel is

$$\kappa_{\varphi_g}(t) = \frac{1}{2\varphi_g} e^{-|t|/\varphi_g}, \quad \widehat{\kappa_{\varphi_g}}(\omega) = \frac{1}{1+\varphi_g^2\omega^2}, \quad \varphi_g = \frac{1+\sqrt{5}}{2}.$$

We write $\Lambda(n)$ for the von Mangoldt function and $\Xi(z) = \xi(\frac{1}{2}+iz)$ for the completed zeta on the critical line. We denote $\mathcal{H}$ the Hilbert space of $-Schmidt$ ideals, respectively.

We use standard facts on spectral multipliers, perturbation determinants, spectral shift, and the de Branges Hermite–Biehler formalism; core points are summarized in the appendices and referenced.

3 Operator construction via golden resonance

We construct a bounded, self-adjoint perturbation K of A_0 by regularized prime sums with the golden kernel, then the resolvent limit to define $A = A_0 + K$.

3.1 Regularized symbols and boundedness

Let $\eta \in C_c^\infty(\mathbb{R})$ be even with $\eta(0) = 1$ and set $\eta_X(u) := \eta(u/X)$ for $X \geq 1$. Define for $\omega \in \mathbb{R}$, $S_X(\omega) := 2 \widehat{\kappa_{\varphi_g}}(\omega) \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \eta_X(\log n) \cos(\omega \log n) + \Gamma\text{-term}_X(\omega)$, (3.1) where $\Gamma\text{-term}_X$ is the standard archimedean multiplier $\mathcal{F}^{-1} M_{S_X} \mathcal{F}$, i.e. the Fourier multiplier by S_X .

Lemma 3.1 (Bounded, self-adjoint perturbations). *For each $X \geq 1$, K_X is bounded and self-adjoint on $L^2(\mathbb{R})$, with $\sup_X \|K_X\| < \infty$.*

Proof. The symbol S_X is real and even, so K_X is self-adjoint. The compact support of $\eta_X \circ \log$ restricts the sum to $(1 + \varphi_g^2 \omega^2)^{-1}$ ensures $S_X \in L^\infty$ uniformly, hence $\|K_X\| \lesssim 1$. $\square$

3.2 Norm-resolvent limit and the operator A

Define $A_X := A_0 + K_X$ on $H^1(\mathbb{R})$. We pass $X \rightarrow \infty$ to obtain A.

Proposition 3.2 (Norm-resolvent convergence). *There exists a bounded, self-adjoint K and self-adjoint $A := A_0 + K$ such that*

$$\lim_{X \rightarrow \infty} \|(A_X - z)^{-1} - (A - z)^{-1}\| = 0 \quad (z \in \mathbb{C} \setminus \mathbb{R}).$$

Moreover, $K_X \rightarrow K$ strongly and $\sup_X \|K_X\| < \infty$.

Proof. Pointwise a.e. convergence $S_X(\omega) \rightarrow S(\omega)$ follows from the explicit formula with the cut off f_{η_X} removed. M_S strongly and uniformly boundedly, hence $K_X \rightarrow K$ strongly. Bounded perturbation preserves self-adjointness (Kato – Rellich), and standard results yield norm – resolvent convergence. $\square$

Remark 3.3. The role of $\widehat{\kappa_{\varphi_g}}$ is twofold : spectral smoothing for boundedness and a rational decay profile that for

4 Trace identity and the explicit formula

We match the Weil explicit formula by testing A against even Schwartz functions with compactly supported Fourier transform.

Definition 4.1. Let $\mathcal{S}_{\text{even},c} = \{\varphi \in \mathcal{S}(\mathbb{R}) : \varphi \text{ even, } \text{supp } \widehat{\varphi} \text{ compact}\}$.

Theorem 4.2 (Trace identity). *For $\varphi \in \mathcal{S}_{\text{even},c}$, $\text{Tr}(\varphi(A) - \varphi(A_0)) = \text{Re}\left(2 \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \widehat{\varphi}(\log n)\right) + \mathcal{A}_{\Gamma}[\varphi]$, (4.1) where $\mathcal{A}_{\Gamma}$ is the standard archimedean term arising from the Γ – factor of ξ .*

Proof. For finite X, spectral multiplier theory with compactly supported $\widehat{\varphi}$ implies $\varphi(A_X) - \varphi(A_0) \in \mathfrak{S}_1$. On the Fourier side, the trace is the integral of $\widehat{\varphi}(\cdot)$ against $S_X(\cdot)$, yielding the prime sum plus gamma ∞ using Theorem 3.2 and dominated convergence for the symbols. $\square$

Corollary 4.3 (Riemann–von Mangoldt). *Let $N_A(T)$ denote the number of eigenvalues of A in $[0, T]$, counting*

$$N_A(T) = \frac{T}{2\pi} \log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + O(\log T).$$

Proof. Apply Theorem 4.2 to a smooth approximation of the indicator of $[-T, T]$ and evaluate the archimedean contribution as usual. $\square$

5 Determinant identity and spectral bijection

Let $\Delta(z)$ be the perturbation determinant of A relative to A_0 , normalized at infinity.

Definition 5.1 (Perturbation determinant). For $z \in \mathbb{C} \setminus \mathbb{R}$, define

$$\Delta(z) := \det \left(\text{Id} + K(A_0 - z)^{-1} \right),$$

with analytic continuation to an entire function controlled by growth estimates (see Appendix C).

Proposition 5.2 (Functional symmetry and growth). $\Delta(z)$ is entire, of finite order and type determined by $\widehat{\kappa_{\varphi_g}}$

$$\overline{\Delta(\bar{z})} = \Delta(z), \quad \Delta(-z) = \Delta(z) e^{p(z)}$$

for some even polynomial p of degree ≤ 2 .

Proof. Reality and evenness of the symbol yield $\overline{\Delta(\bar{z})} = \Delta(z)$. The rational decay $(1 + \varphi_g^2 \omega^2)^{-1}$ fixes the order, h
2. Evenness stems from the kernel symmetry. $\square$

Theorem 5.3 (Hadamard locking to Ξ). There exist $C \neq 0$ and an even polynomial q of degree ≤ 2 such that

$$\Delta(z) = C e^{q(z)} \Xi(z), \quad \Xi(z) := \xi\left(\frac{1}{2} + iz\right).$$

Proof. By Theorem 4.3, Δ and Ξ have matching zero-counting asymptotics. The trace identity (4.1) aligns the h
2; normalization at infinity fixes C, q . $\square$

Corollary 5.4 (Spectral bijection). The zeros of Ξ coincide with the spectrum of A (counting multiplicity).

Proof. Zeros of Δ are poles of the resolvent and hence eigenvalues of A . Apply Theorem 5.3. $\square$

6 de Branges positivity and the critical line

Define the transfer function

$$S(z) := \Delta(-z) \frac{1}{\Delta(z)}, \quad m(z) := i \frac{1 + S(z)}{1 - S(z)}.$$

By Theorem 5.2, S is Schur-class in $\text{Im } z > 0$ and is a Herglotz function.

Proposition 6.1 (Hermite–Biehler structure). There exists a Hermite–Biehler entire function E such that

$$S(z) = E^\#(z) \frac{1}{E(z)}, \quad E^\#(z) := \overline{E(\bar{z})},$$

and

$$\Xi(z) \propto E(z) E^\#(z).$$

The associated de Branges space $H(E)$ is well-defined with its canonical reproducing kernel.

Proof. Schur/Herglotz factorization produces E with no zeros in $\text{Im } z > 0$ and $S = E^\# / E$. The relation with Ξ follows from Proposition 5.3.

Main Theorem 1 (Positivity forces the critical line). All zeros of Ξ are real. Equivalently, all nontrivial zeros of ζ lie on the critical line $\frac{1}{2}$.

Proof. In $H(E)$, positivity of the reproducing kernel together with the Hermite–Biehler property implies that the zeros of $E(z)E^\#(z)$ lie on $\mathbb{R}$. Since $\Xi \propto EE^\#$, all zeros of Ξ are real. $\square$

Remark 6.2. The golden kernel fixes both the determinant’s growth (hence genus/type and the allowed polynomial q) and the Schur/Herglotz class of S . This dual control welds arithmetic (trace) and analysis (positivity) into a single invariant choice, removing tuning freedom.

7 Conclusion

We presented a canonical operator framework anchored by a golden-ratio resonance kernel, delivering (i) an explicit-trace identity, (ii) a determinant identity $\Delta = Ce^q \Xi$ that pins the spectrum to zero, and line localization. The architecture minimizes arbitrariness: the single geometric choice φ_g dictates decay, growth, and localization.

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A Kernel estimates and bounded multipliers

Lemma A.1. Let $\widehat{\kappa_{\varphi_g}}(\omega) = (1 + \varphi_g^2 \omega^2)^{-1}$. For any even cutoff $f \eta \in C_c^\infty(\mathbb{R})$ and $X \geq 1$, the symbol S_X in (3.1) is real and bounded by ∞ .

Proof. Compact support of $u \mapsto \eta(u/X)$ restricts $\log n$ to a bounded window. In that window, the weighted sum $\sum \Lambda(n) n^{-1/2} \widehat{\varphi}(\log n)$ is bounded.

B The explicit formula via trace identities

For $\varphi \in \mathcal{S}_{\text{even},c}$, $\varphi(A_X) - \varphi(A_0) \in \mathfrak{S}_1$ and

$$\text{Tr} \left(\varphi(A_X) - \varphi(A_0) \right) = \int_{\mathbb{R}} \widehat{\varphi}(\omega) S_X(\omega) \frac{d\omega}{2\pi}.$$

As $X \rightarrow \infty$, the prime sum term converges to $2 \text{Re} \sum \Lambda(n) n^{-1/2} \widehat{\varphi}(\log n)$; the archimedean term contributes $A_\Gamma[\varphi]$ by resolvent convergence (Theorem 3.2) and dominated convergence justifies the limit, yielding (4.1).

C Determinants, Hadamard factorization, and growth

For bounded self-adjoint K , $\text{Id} + K(A_0 - z)^{-1}$ is a trace-class perturbation of Id for $z \notin \mathbb{R}$. The perturbation determinant $\Delta(z)$ extends to an entire function controlled by the decay of $\widehat{\kappa_{\varphi_g}}$. Using Theorem 4.2 and Theorem 4.3, one identifies $\Delta(z)e^{p(z)}$ stems from kernel evenness; the polynomial p is constrained by order/type. Hadamard’s factorization $\Delta(z) = Ce^q \Xi$ with even $\deg q \leq 2$. Normalization at infinity fixes C, q .

D Scattering, spectral shift, and Birman–Kreĭn

Let $\xi(\lambda)$ be the spectral shift function for (A, A_0) . Then

$$\det \left(\text{Id} + K(A_0 - (\lambda + i0))^{-1} \right) = \exp \left(-i\pi \xi(\lambda) \right).$$

Differentiating relates $\xi'(\lambda)$ to the explicit–formula density, matching the zero distribution of Ξ and providing

E de Branges spaces and positivity

Given Schur S in $\text{Im } z > 0$, the Herglotz transform $m(z) = i \frac{1+S}{1-S}$ admits a canonical representation. There exists a de Branges space $\mathcal{H}(E)$ with $S = E^\# / E$. The de Branges space $\mathcal{H}(E)$ consists of entire F with

$$\|F\|_{\mathcal{H}(E)}^2 = \int_{\mathbb{R}} \frac{|F(x)|^2}{|E(x)|^2} dx < \infty,$$

and reproducing kernel

$$K(z, w) = E(z)E^\#(\bar{w}) - E^\#(z)E(\bar{w}) \frac{1}{2\pi i(\bar{w} - z)}.$$

Positivity of K forces zeros of $EE^\#$ to lie on $\mathbb{R}$, implying $\Xi \propto EE^\#$.

References

- [1] L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice-Hall, 1968.
- [2] E.C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D.R. Heath-Brown, Oxford Univ. Press, 1986.
- [3] H.M. Edwards, *Riemann’s Zeta Function*, Dover, 2001.
- [4] B. Simon, *Trace Ideals and Their Applications*, 2nd ed., AMS, 2005.
- [5] D.R. Yafaev, *Mathematical Scattering Theory*, AMS, 1992.
- [6] M.S. Birman and M.G. Kreĭn, *On the theory of wave operators and scattering operators*, Dokl. Akad. Nauk SSSR 144 (1962), 475–478.
- [7] J.B. Conrey, *The Riemann Hypothesis*, Notices of the AMS 50 (2003), no. 3, 341–353.